Pricing Arithmetic Asian Options via Monte Carlo Simulation and Control Variate Variance Reduction

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Abstract

This paper presents a mathematical and numerical framework for pricing discrete arithmetic Asian options within the Black-Scholes-Merton model. Due to the absence of a closed-form analytical distribution for the arithmetic average of a geometric Brownian motion, we implement a standard Monte Carlo estimator and evaluate two convergence acceleration techniques under equal computational budgets: antithetic variates and a control variate method leveraging the exact analytical solution of the discrete geometric Asian option. Using an out-of-sample pilot scheme to eliminate finite-sample bias, numerical experiments demonstrate an empirical correlation of $\hat{\rho} \approx 0.9996$, yielding a variance reduction factor exceeding $1,300\times$ and a net CPU time-adjusted efficiency gain above $900\times$.

1 Probabilistic Framework and Underlying Asset Dynamics

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, where $(\mathcal{F}_t)_{t\in[0,T]}$ is the natural filtration generated by a standard one-dimensional Brownian motion $(W_t)_{t\in[0,T]}$.

By Girsanov's theorem and the completeness of the Black-Scholes market (Second Fundamental Theorem of Asset Pricing), there exists a unique equivalent martingale measure $\mathbb{Q}$ under which the discounted price process is a martingale. Under $\mathbb{Q}$, the dynamics of the non-dividend-paying underlying asset $(S_t)_{t\in[0,T]}$ are governed by the linear stochastic differential equation (SDE):

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^{\mathbb{Q}}, \quad S_0 > 0 \tag{1}$$

where $r \in \mathbb{R}$ denotes the constant continuous risk-free interest rate, $\sigma > 0$ is the constant volatility, and $(W_t^{\mathbb{Q}})_{t\in[0,T]}$ is a standard $\mathbb{Q}$-Brownian motion.

Proposition 1 (Strong Solution and Exact Discretization)**.** *Because the drift and diffusion coefficients satisfy global Lipschitz and linear growth conditions, the SDE admits a unique strong solution. Applying Itô's Lemma to $f(x) = \ln(x)$ yields the explicit form for all $t \in [0, T]$:*

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma W_t^{\mathbb{Q}}\right) \tag{2}$$

Over a uniform monitoring grid $0 = t_0 < t_1 < \cdots < t_m = T$ with time step $\Delta t = T/m$, the exact transition law satisfies the recursive scheme:

$$S_{t_j} = S_{t_{j-1}} \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}Z_j\right), \quad \text{where } (Z_j)_{1\le j\le m} \underset{i.i.d.}{\sim} \mathcal{N}(0,1) \tag{3}$$

2 Asian Options and the Valuation Problem

Unlike standard European vanilla options, the payoff of an Asian option depends on the average price of the underlying asset over the monitoring period $[0, T]$. This path-dependent averaging reduces sensitivity to spot price manipulation near maturity and yields a lower effective volatility than the underlying asset itself, making the Asian option less expensive than its European counterpart.

Definition 1 (Discrete Arithmetic Asian Call)**.** *Given a monitoring grid $(t_1, \dots, t_m)$, the terminal payoff at maturity T of a discrete arithmetic Asian Call with strike $K > 0$ is defined as:*

$$H_A = (A_T - K)_+ = \max\left(\frac{1}{m}\sum_{j=1}^{m} S_{t_j} - K,\, 0\right) \tag{4}$$

The arbitrage-free price at $t = 0$ is the discounted expectation under $\mathbb{Q}$:

$$C_A = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(A_T - K)_+\right] \tag{5}$$

Since a sum of log-normal random variables does not follow a known closed-form distribution, C_A cannot be evaluated analytically. Conversely, the discrete geometric average preserves log-normality.

Proposition 2 (Closed-Form Solution for the Discrete Geometric Asian Call)**.** *Let $G_T = \left(\prod_{j=1}^{m} S_{t_j}\right)^{\frac{1}{m}}$ denote the discrete geometric average. The arbitrage-free price at $t = 0$ of the geometric Asian Call, $C_G = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(G_T - K)_+]$, admits an exact analytical formula:*

$$C_G = e^{-rT}\left(S_0 e^{\widetilde{r}T}\Phi(d_1) - K\Phi(d_2)\right) \tag{6}$$

where Φ is the standard normal cumulative distribution function, and the effective discrete parameters are given by:

$$\widetilde{\sigma} = \sigma\sqrt{\frac{(m+1)(2m+1)}{6m^2}}, \qquad \widetilde{r} = \left(r - \frac{\sigma^2}{2}\right)\frac{m+1}{2m} + \frac{\widetilde{\sigma}^2}{2}$$

$$d_1 = \frac{\ln(S_0/K) + (\widetilde{r} + \frac{1}{2}\widetilde{\sigma}^2)T}{\widetilde{\sigma}\sqrt{T}}, \qquad d_2 = d_1 - \widetilde{\sigma}\sqrt{T}$$

Proof. Expanding the log-geometric average yields:

$$\ln\left(\frac{G_T}{S_0}\right) = \frac{1}{m}\sum_{j=1}^{m}\ln\left(\frac{S_{t_j}}{S_0}\right) = \left(r - \frac{\sigma^2}{2}\right)\frac{1}{m}\sum_{j=1}^{m} t_j + \frac{\sigma}{m}\sum_{j=1}^{m} W_{t_j}^{\mathbb{Q}}$$

Since $t_j = j\frac{T}{m}$, the deterministic time average simplifies to $\frac{T}{m^2}\frac{m(m+1)}{2} = T\frac{m+1}{2m}$. Expressing $W_{t_j}^{\mathbb{Q}} = \sum_{k=1}^{j}(W_{t_k}^{\mathbb{Q}} - W_{t_{k-1}}^{\mathbb{Q}})$ allows us to rewrite the Brownian sum in terms of independent increments:

$$\frac{\sigma}{m}\sum_{j=1}^{m} W_{t_j}^{\mathbb{Q}} = \frac{\sigma}{m}\sum_{k=1}^{m}(m-k+1)(W_{t_k}^{\mathbb{Q}} - W_{t_{k-1}}^{\mathbb{Q}})$$

By the independence and stationarity of Brownian increments, this linear combination is normally distributed with zero mean and variance:

$$\frac{\sigma^2}{m^2}\frac{T}{m}\sum_{u=1}^{m} u^2 = \sigma^2 T\frac{m(m+1)(2m+1)}{6m^3} = \widetilde{\sigma}^2 T$$

Hence, $\ln(G_T/S_0) \sim \mathcal{N}\left((\widetilde{r} - \frac{1}{2}\widetilde{\sigma}^2)T,\, \widetilde{\sigma}^2 T\right)$, reducing the expectation to a Black-Scholes-type valuation with effective drift $\widetilde{r}$ and volatility $\widetilde{\sigma}$. □

3 Monte Carlo Estimators and Variance Reduction

3.1 Standard Estimator and Convergence Rate

Let $(Y^{(i)})_{1\leq i\leq N}$ be i.i.d. replications of the discounted arithmetic payoff $Y = e^{-rT}(A_T - K)_+$. By the Strong Law of Large Numbers, the sample mean $\hat{C}_N^{\mathrm{MC}} = \frac{1}{N}\sum_{i=1}^{N} Y^{(i)}$ converges almost surely to C_A. By the Central Limit Theorem (given $\mathbb{V}\mathrm{ar}(Y) = \sigma_Y^2 < +\infty$):

$$\sqrt{N}\left(\hat{C}_N^{\mathrm{MC}} - C_A\right) \xrightarrow[N\to+\infty]{\mathcal{L}} \mathcal{N}(0, \sigma_Y^2) \tag{7}$$

The standard error decays at rate $\mathcal{O}(N^{-1/2})$.

3.2 Antithetic Variates at Equal Computational Budget

By symmetry of the standard normal distribution, if $Z = (Z_1, \ldots, Z_m) \sim \mathcal{N}(0, I_m)$, then $-Z \sim \mathcal{N}(0, I_m)$. To ensure a fair comparison at an equal computational budget of N total path evaluations (assuming N is even), we generate $N/2$ independent pairs:

$$\hat{C}_{N/2}^{\mathrm{Anti}} = \frac{1}{N/2}\sum_{i=1}^{N/2} \frac{Y(Z^{(i)}) + Y(-Z^{(i)})}{2} \tag{8}$$

Because the payoff mapping $Z \mapsto Y(Z)$ is coordinate-wise non-decreasing and $Z \mapsto Y(-Z)$ is coordinate-wise non-increasing in the independent Gaussian shocks Z_j, Harris's inequality guarantees that $\mathbb{C}\mathrm{ov}(Y(Z), Y(-Z)) \leq 0$. Consequently, the variance at equal computational cost satisfies:

$$\mathbb{V}\mathrm{ar}\left(\hat{C}_{N/2}^{\mathrm{Anti}}\right) = \frac{\mathbb{V}\mathrm{ar}(Y)}{N}\left(1 + \rho_{Y(Z),Y(-Z)}\right) \leq \mathbb{V}\mathrm{ar}\left(\hat{C}_N^{\mathrm{MC}}\right) \tag{9}$$

3.3 Control Variate Method and Pilot Estimation

We exploit the strong collinearity between the discounted arithmetic payoff $Y = e^{-rT}(A_T - K)_+$ and the discounted geometric payoff $X = e^{-rT}(G_T - K)_+$, whose exact expectation $\mathbb{E}^{\mathbb{Q}}[X] = C_G$ is known from Proposition 2.

For any fixed coefficient $\beta \in \mathbb{R}$, the control variate estimator is defined as:

$$\hat{C}_N^{\mathrm{CV}}(\beta) = \frac{1}{N}\sum_{i=1}^{N}\left(Y^{(i)} - \beta\left(X^{(i)} - C_G\right)\right) = \bar{Y}_N - \beta\left(\bar{X}_N - C_G\right) \tag{10}$$

Proposition 3 (Optimal Control Variate)**.** *For any deterministic $\beta \in \mathbb{R}$, $\hat{C}_N^{CV}(\beta)$ is an unbiased estimator of C_A. Its variance is uniquely minimized at:*

$$\beta^* = \frac{\mathbb{C}ov(X, Y)}{\mathbb{V}ar(X)} \tag{11}$$

yielding a theoretical variance reduction ratio of:

$$\frac{\mathbb{V}ar\left(\hat{C}_N^{CV}(\beta^*)\right)}{\mathbb{V}ar\left(\hat{C}_N^{MC}\right)} = 1 - \rho_{X,Y}^2 \tag{12}$$

where $\rho_{X,Y} \in [-1, 1]$ denotes the Pearson correlation coefficient between X and Y.

Proof. By linearity of expectation, $\mathbb{E}^{\mathbb{Q}}[\hat{C}_N^{\mathrm{CV}}(\beta)] = \mathbb{E}^{\mathbb{Q}}[Y] - \beta(\mathbb{E}^{\mathbb{Q}}[X] - C_G) = C_A$. The single-path variance $g(\beta) = \mathbb{V}\mathrm{ar}(Y - \beta X) = \mathbb{V}\mathrm{ar}(Y) - 2\beta\mathbb{C}\mathrm{ov}(X, Y) + \beta^2\mathbb{V}\mathrm{ar}(X)$ is strictly convex in β since $\mathbb{V}\mathrm{ar}(X) > 0$, attaining its global minimum at $g'(\beta^*) = 0 \iff \beta^* = \frac{\mathbb{C}\mathrm{ov}(X,Y)}{\mathbb{V}\mathrm{ar}(X)}$. Substituting β^* gives $\mathbb{V}\mathrm{ar}(Y - \beta^* X) = \mathbb{V}\mathrm{ar}(Y)(1 - \rho_{X,Y}^2)$. □

Remark 1 (Out-of-Sample Pilot Estimation)**.** *Estimating $\hat{\beta}^*$ and evaluating $\hat{C}_N^{CV}$ on the same sample introduces a finite-sample bias of order $\mathcal{O}(1/N)$. To maintain strict unbiasedness, we estimate $\hat{\beta}^*$ on an independent pilot run of size $N_{pilot} = 10,000$ prior to computing the estimator on the $N = 100,000$ primary trajectories. The standard error is computed from the sample variance of the controlled observations $Y^{(i)} - \hat{\beta}^* X^{(i)}$.*

4 Model Validation and Numerical Results

Simulations are executed in Python (`NumPy PCG64`, fixed seed `42`) with parameters: $S_0 = 100$, $K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1.0$ year, and $m = 252$ daily steps.

4.1 Validation Benchmarks

1. **Boundary Consistency ($m = 1$):** For $m = 1$, the Asian option collapses to a standard European Call. Numerically, $C_G(m = 1) = 10.450584$ matches the exact Black-Scholes price to machine precision, with zero control variate variance.

2. **Discrete Geometric Benchmark ($m = 252$):** Proposition 2 yields $\widetilde{\sigma} = 0.11581$, $\widetilde{r} = 0.02177$, $d_1 = 0.2458$, and $d_2 = 0.1300$, resulting in an exact analytical price of $\mathbf{C_G = 5.5655}$ (differing from 5.5468 in the continuous $m \to \infty$ limit).

3. **Analytical Approximations & Large-Scale MC:** The Turnbull-Wakeman log-normal moment-matching benchmark gives an order-of-magnitude estimate of $C_{\text{TW}} \approx 5.801$. A high-precision reference Monte Carlo simulation ($N = 10^7$) independently confirms the benchmark price at $\mathbf{5.7818 \pm 0.0002}$.

4.2 Empirical Performance Comparison

The independent pilot run ($N_{\text{pilot}} = 10,000$) yields $\hat{\beta}^* = 1.0354$. Across the $N = 100,000$ out-of-sample paths, the empirical correlation is $\hat{\rho}_{X,Y} = 0.999621$, matching the theoretical variance reduction factor $1/(1 - \hat{\rho}_{X,Y}^2) \approx 1,319\times$.

Table 1: Estimator performance at equal budget ($N = 100,000$ total path evaluations)

Estimation Method	Estimated Price	Standard Error	95% Confidence Interval	Equal-Budget VRF	CPU Time (s)	Time-Adjusted Gain
Standard MC (N paths)	5.7888	0.0254	[5.7391, 5.8385]	1.0×	0.68 s	1.0×
Antithetic ($N/2$ pairs)	5.7924	0.0178	[5.7575, 5.8273]	2.0×	0.70 s	1.94×
Control Variate (N paths)	**5.7821**	**0.0007**	**[5.7807, 5.7835]**	**1,315.2×**	**0.99 s**	**903.6×**

Table 1 and Figure 1 demonstrate consistent alignment:

- The control variate estimate (5.7821 ± 0.0007) falls cleanly within the high-precision reference interval (5.7818 ± 0.0002).
- Antithetic variates at equal budget achieve a 2.0× variance reduction and a 1.94× net CPU gain.
- Accounting for pilot and geometric average overhead, the control variate method achieves a net time-adjusted efficiency gain exceeding 900×.

5 Conclusion

This study illustrates how leveraging an exact closed-form solution of an auxiliary contract (the discrete geometric Asian Call) as a control variate overcomes the slow convergence of Monte Carlo simulation for path-dependent derivatives. Future extensions include combining antithetic

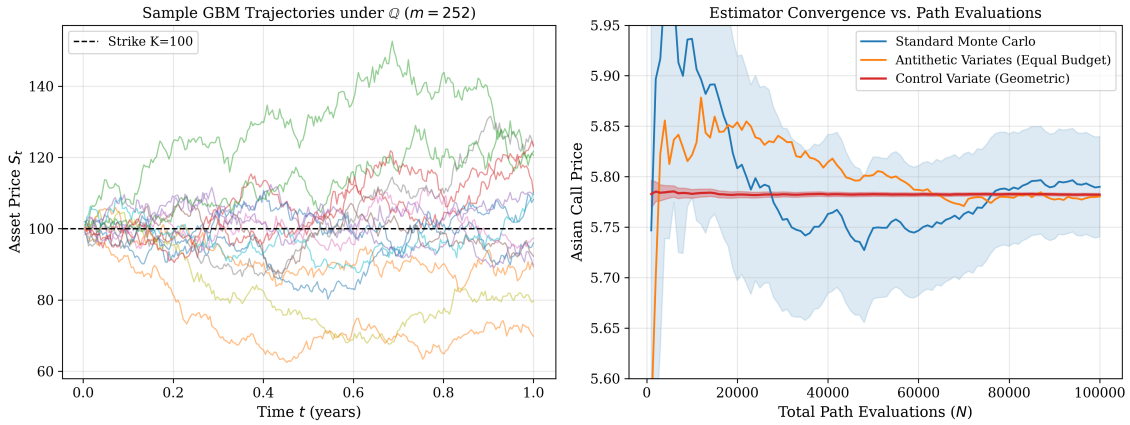


Figure 1: Left: Sample trajectories of the geometric Brownian motion under $\mathbb{Q}$. Right: Convergence of the three estimators as a function of total path evaluations N towards the true price (≈ 5.782), showing the sharp contraction of the 95% confidence interval.

sampling with multiple control variates (e.g., incorporating the terminal spot S_T), computing pathwise sensitivities (Delta Δ, Vega $\mathcal{V}$), and extending the framework to stochastic volatility models.